Composite Integer Parts of Powers of Algebraic Units

Pure Cubic Fields, Salem Numbers, and a K3 Application

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Abstract

We study algebraic units whose power sequences have composite integer parts. For every cubefree integer $m > 1$, every unit $\varepsilon > 1$ in $\mathbb{Q}(\sqrt[3]{m})$, and every $n \geq 1$, we prove that $\lfloor \varepsilon^{6n} \rfloor$ is congruent to 2 or 3 modulo 6 and is composite. The proof uses the discriminant square class -3 to restrict factorization modulo 2 and 3. An elementary norm estimate removes all initial exceptions without invoking the least-Pisot-number theorem. We also prove that every Salem number τ has an effectively computable positive power α for which $\lfloor \alpha^n \rfloor$ is composite for every $n \geq 1$. A norm estimate confines the difference between the trace and the integer part to a finite window, and a finite set of primes covers the moving residue. An exact quartic example admits exponent 260. Finally, Brandhorst's stable realization theorem yields projective K3 automorphisms whose dynamical degrees satisfy the same compositeness property. The K3 assertion concerns powers and is a corollary, not a classification of individual dynamical degrees.

Keywords. Pisot number; Salem number; algebraic unit; trace congruence; composite integer part; K3 surface.

1 Introduction and principal results

For a real number $\alpha > 1$, consider the integer sequence

$$a_n(\alpha) = \lfloor \alpha^n \rfloor, \qquad n \geq 1.$$

Compositeness along this sequence can arise from an arithmetic property of a nearby integer trace. If all other conjugates are small, that trace is particularly close to α^n. If some conjugates lie on the unit circle, the error need not converge to zero; nevertheless, a bounded error can still suffice when its possible integer values are covered by several small prime divisors.

Hahn, Ismailescu, Kim and Kim [1] give families of algebraic integers with eventually composite power floors. Their Conjecture 1 concerns the sixth powers of fundamental units in pure cubic fields. Their final section also asks for an explicit non-Pisot example with eventually composite power floors. The two results below address these particular statements, with an all-index conclusion. Statements about their mathematical implications are separate from claims of historical priority; this manuscript makes no assertion that a literature search proves the absence of earlier equivalent results.

Theorem 1.1 (Pure cubic units; source T54)**.** *Let $m > 1$ be a cubefree positive integer, let $K = \mathbb{Q}(\sqrt[3]{m})$, and let $\varepsilon \in \mathcal{O}_K^\times$ satisfy $\varepsilon > 1$ in the real embedding. Then, for every integer $n \geq 1$,*

$$\left\lfloor \varepsilon^{6n} \right\rfloor \equiv 2 \text{ or } 3 \pmod 6.$$

Every integer in this sequence is composite.

In particular, taking ε to be the fundamental unit proves the statement of [1, Conjecture 1], and the conclusion holds from $n = 1$. The extension from a fundamental unit to all positive units is not counted as an independent phenomenon: the positive unit group here has rank one.

Theorem 1.2 (Salem power cover; source T60)**.** *Let τ be a Salem number of degree $d \geq 4$. There is an effectively computable integer $h \geq 1$ such that*

$$\left\lfloor \tau^{hn} \right\rfloor \text{ is composite for every } n \geq 1.$$

The number τ^h is itself a Salem number of degree d. Each displayed integer has a prime divisor at most $2d - 3$.

Here a Salem number has one real conjugate greater than one, its reciprocal, and all remaining conjugates on the unit circle; degree two is excluded in our convention. Thus Theorem 1.2 gives a non-Pisot example in the sense of [1, Section 7]. It supplies neither a rational noninteger example nor a transcendental example.

The common mechanism is elementary once suitable error control is established. We first state it separately, then prove the cubic and Salem estimates. A final geometric application uses an established existence theorem and the elementary behavior of dynamical degree under iteration.

2 Trace corrections and finite prime covers

Let $\alpha > 1$ be an algebraic integer of degree d, with conjugates $\alpha = \alpha_1, \ldots, \alpha_d$. Write

$$T_r = \mathrm{Tr}_{\mathbb{Q}(\alpha)/\mathbb{Q}}(\alpha^r) = \sum_{j=1}^{d} \alpha_j^r, \qquad E_r = \sum_{j=2}^{d} \alpha_j^r.$$

Then $T_r \in \mathbb{Z}$ and E_r is real. There is an exact identity, including cases where E_r happens to be an integer:

$$\lfloor \alpha^r \rfloor = T_r - \lceil E_r \rceil \,. \tag{1}$$

Indeed, for every integer T and real x, $\lfloor T - x \rfloor = T - \lceil x \rceil$.

Lemma 2.1 (Finite-cover criterion)**.** *Let $M \geq 2$, $t \in \mathbb{Z}$, and $Q \subset \mathbb{Z}$ be finite. Suppose that for every $n \geq 1$ an exponent r_n satisfies*

$$T_{r_n} \equiv t \pmod{M}, \qquad \lceil E_{r_n} \rceil \in Q.$$

Suppose that each integer $t - q$, $q \in Q$, has a prime divisor $p_q \mid M$, and that

$$\lfloor \alpha^{r_n} \rfloor > \max_{q \in Q} p_q \qquad (n \geq 1).$$

Then every $\lfloor \alpha^{r_n} \rfloor$ is composite.

Proof. For the actual value $q = \lceil E_{r_n} \rceil$, equation (1) gives $\lfloor \alpha^{r_n} \rfloor \equiv t - q \pmod{p_q}$. Thus p_q divides the floor. The strict size inequality makes this a proper prime divisor. □

The prime is allowed to depend on n. This distinction is essential for the Salem argument: a uniformly bounded correction does not generally identify one prime dividing all terms. No probabilistic statement about primality enters the criterion.

Lemma 2.2 (Residue-field power sums)**.** *Let $F \in \mathbb{Z}[X]$ be monic of degree d and let $p \nmid F(0)$. If all roots of its reduction modulo p, counted with multiplicity, belong to $\mathbb{F}_{p^e}$, then its Nth power sum satisfies*

$$s_N(F) \equiv d \pmod{p} \qquad \text{whenever } p^e - 1 \mid N.$$

Proof. Every reduced root is nonzero and therefore has multiplicative order dividing $p^e - 1$. Its Nth power is one. Reduction of the integral power sum equals the sum of these powers with their multiplicities. Repeated roots cause no difficulty. □

3 The arithmetic of pure cubic units

3.1 Signature, norm, and discriminant square class

Set $r = \sqrt[3]{m}$. Since m is cubefree and greater than one, $X^3 - m$ is irreducible over $\mathbb{Q}$ and K has signature $(1, 1)$. A unit $\varepsilon > 1$ cannot be rational, because the only rational algebraic-integer units are ± 1. Since $[K : \mathbb{Q}] = 3$, it follows that $K = \mathbb{Q}(\varepsilon)$.

Let $\rho, \overline{\rho}$ be its nonreal conjugates. Its norm is positive and is an integer unit, whence

$$\mathrm{N}_{K/\mathbb{Q}}(\varepsilon) = \varepsilon|\rho|^2 = 1, \qquad |\rho| = \varepsilon^{-1/2}. \tag{2}$$

In particular, ε is a cubic Pisot number. Its minimal polynomial has the form

$$F(X) = X^3 - aX^2 + bX - 1, \qquad a, b \in \mathbb{Z}. \tag{3}$$

Lemma 3.1. *The discriminant of F equals $-3q^2$ for an integer $q \neq 0$.*

Proof. The discriminant of the basis $(1, r, r^2)$ is $-27m^2$. The basis $(1, \varepsilon, \varepsilon^2)$ is related to it by an invertible rational matrix, so its discriminant is $-27m^2u^2 = -3(3mu)^2$ for some $u \in \mathbb{Q}^\times$. This is also the polynomial discriminant of F and is an integer. Write $3mu = v/w$ in lowest terms. Integrality implies $w^2 \mid 3$, hence $w = 1$. Thus $q = v \in \mathbb{Z} \setminus \{0\}$. □

This argument avoids a case distinction for integral bases when $m \equiv \pm 1 \pmod 9$. It also avoids an assumption that $\mathbb{Z}[\varepsilon]$ is the full ring of integers. All possible basis indices are absorbed by the discriminant square class.

3.2 Local factorization at two and three

The polynomial discriminant in the variables of (3) is

$$\mathrm{disc}(F) = a^2b^2 - 4b^3 - 4a^3 - 27 + 18ab. \tag{4}$$

Lemma 3.2. *The reduction of F modulo 2 is either*

$$(X + 1)^3 \qquad \text{or} \qquad (X + 1)(X^2 + X + 1).$$

Consequently $\mathrm{Tr}(\varepsilon^{6n}) \equiv 1 \pmod 2$ *for all* $n \geq 1$.

Proof. The two irreducible monic cubics over $\mathbb{F}_2$ with constant term one are $X^3 + X + 1$ and $X^3 + X^2 + 1$. They correspond to a, b of opposite parity. In either case, direct reduction of (4) gives

$$\mathrm{disc}(F) \equiv 1 \pmod 8.$$

For completeness, if a is even and b is odd, $a^2+2ab \equiv 0 \pmod 8$ and $-4b^3-27 \equiv 1 \pmod 8$. If a is odd and b is even, $b^2+2ab \equiv 0 \pmod 8$ and $-4a^3-27 \equiv 1 \pmod 8$. On the other hand, $-3q^2$ modulo 8 belongs to $\{0,4,5\}$. Irreducibility is therefore impossible. The two remaining monic cubics with constant term one are precisely the displayed polynomials. Their roots are in $\mathbb{F}_4^\times$, and Lemma 2.2 applies because $3 \mid 6n$. □

Lemma 3.3. *The reduction of F modulo 3 is either*

$$(X-1)^3 \qquad \text{or} \qquad (X+1)^2(X-1).$$

Consequently $\mathrm{Tr}(\varepsilon^{6n}) \equiv 0 \pmod 3$ *for all* $n \geq 1$.

Proof. The discriminant vanishes modulo 3, so F has a repeated factor. If $F' = 0$, then $F = X^3 - 1 = (X-1)^3$. Otherwise a repeated irreducible factor has degree one, since twice its degree is at most three; the remaining factor also has degree one. All roots therefore lie in $\mathbb{F}_3^\times = \{1,-1\}$. Their product is one, so either all three are one or exactly two are minus one. Their even powers are all one, and the power sum is $3 = 0$ in $\mathbb{F}_3$. □

Combining the two congruences yields

$$\mathrm{Tr}(\varepsilon^{6n}) \equiv 3 \pmod 6 \qquad (n \geq 1). \tag{5}$$

The local argument uses repeated roots with multiplicity; an unramified hypothesis at 3 would be both unnecessary and incorrect here.

4 Removing every initial exception

The trace congruence alone gives eventual information because the other two conjugates decay. To start at $n = 1$, we need a uniform numerical inequality. A simple integer norm is sufficient.

Lemma 4.1 (An elementary lower bound)**.** *Every unit $\varepsilon > 1$ generating a cubic field of signature $(1,1)$ satisfies $\varepsilon > 127/100$. In particular, $\varepsilon^3 > 2$.*

Proof. By (2),

$$1 \leq |\,\mathrm{N}(1-\varepsilon)| = (\varepsilon-1)|1-\rho|^2 \leq (\varepsilon-1)(1+\varepsilon^{-1/2})^2.$$

The norm is a nonzero integer. Suppose first that $1 < \varepsilon \leq 5/4$. Then the last expression is strictly less than $4(\varepsilon-1) \leq 1$, a contradiction. If instead $5/4 < \varepsilon \leq 127/100$, then

$$\varepsilon^{-1/2} < \frac{2}{\sqrt{5}} < \frac{9}{10}, \qquad (\varepsilon-1)(1+\varepsilon^{-1/2})^2 < \frac{27}{100}\left(\frac{19}{10}\right)^2 = \frac{9747}{10000} < 1.$$

This is again impossible. Finally $(127/100)^3 = 2048383/1000000 > 2$. □

Proof of Theorem 1.1. Let $U_n = \mathrm{Tr}(\varepsilon^{6n})$. Equation (2) and Lemma 4.1 imply

$$|\rho^{6n} + \overline{\rho}^{6n}| \leq 2\varepsilon^{-3n} < 1 \qquad (n \geq 1).$$

Thus (1) gives $\lfloor \varepsilon^{6n} \rfloor \in \{U_n - 1, U_n\}$. This statement remains valid if the conjugate sum vanishes. By (5), the possible residues are 2 and 3 modulo 6. Also $\varepsilon^{6n} > 4$, so the floor is at least 4. If its residue is 2 it has proper divisor 2; if its residue is 3 it has proper divisor 3. Hence it is composite. □

Corollary 4.2. *For ε as in Theorem 1.1 and any integer $j \geq 1$, set $\beta = \varepsilon^{2j}$. Then $\lfloor \beta^{3^r} \rfloor$ is composite for every $r \geq 1$.*

Proof. The exponent equals $2j3^r = 6(j3^{r-1})$, so Theorem 1.1 applies without a change in the underlying integer-part operation. □

Example 4.3. Let $r = \sqrt[3]{2}$ and $\varepsilon = 1 + r + r^2$. Since $(r-1)(1+r+r^2) = 1$, this is a unit. Its minimal polynomial is $X^3 - 3X^2 - 3X - 1$. The trace sequence $V_k = \mathrm{Tr}(\varepsilon^k)$ is specified by

$$V_0 = 3, \quad V_1 = 3, \quad V_2 = 15, \qquad V_k = 3V_{k-1} + 3V_{k-2} + V_{k-3}.$$

The first three sixth-power floors are 3243, 10517258, and 34107811455, with residues $3, 2, 3$ modulo 6. These numerical illustrations are not used to establish the theorem; its proof simultaneously treats all indices and all admissible fields.

5 A norm gap for Salem traces

Let τ be Salem of degree $d = 2s + 2$, $s \geq 1$. Write its conjugates as

$$\tau,\ \tau^{-1},\ \zeta_1,\ \overline{\zeta}_1, \ldots, \zeta_s,\ \overline{\zeta}_s, \qquad |\zeta_j| = 1.$$

No ζ_j is a root of unity: otherwise the irreducible minimal polynomial of τ would divide a cyclotomic polynomial, forcing τ itself to be a root of unity.

Lemma 5.1 (Strict correction window)**.** *For every $r \geq 1$,*

$$-(d-2) < E_r := \tau^{-r} + \sum_{j=1}^{s} (\zeta_j^r + \overline{\zeta}_j^r) < d - 2.$$

Consequently,

$$d - \lceil E_r \rceil \in \{2, 3, \ldots, 2d-3\}.$$

Proof. Set $a = \tau^{-r} \in (0,1)$ and $x_j = |1 - \zeta_j^r|^2 > 0$. The norm of the nonzero algebraic integer $\tau^r - 1$ gives

$$1 \leq |\,\mathrm{N}(\tau^r - 1)| = \frac{(1-a)^2}{a} \prod_{j=1}^{s} x_j.$$

Hence $\prod_j x_j \geq a/(1-a)^2 > a$. If $\sum_j x_j \leq a$, the arithmetic–geometric mean inequality would give

$$\prod_j x_j \leq (a/s)^s \leq a,$$

in contradiction to the previous bound. Thus $\sum_j x_j > a$. Since

$$E_r = a + 2s - \sum_j x_j,$$

we obtain $E_r < 2s = d - 2$. The lower bound follows directly from $E_r \geq a - 2s > -(d-2)$. Taking ceilings gives $\lceil E_r \rceil \in \{-d+3, \ldots, d-2\}$, which is the asserted integer window. □

The strict upper endpoint is the role of the norm estimate. Bounding each unit-circle conjugate separately only gives $E_r \leq d - 2 + \tau^{-r}$, which could permit a residual integer equal to one. Such a residue has no prime divisor, so the naive bound is insufficient for the next construction.

6 Effective finite-ring return and the Salem theorem

Let P be the minimal polynomial of τ and let C be its integral companion matrix. Its determinant is one because P is monic reciprocal of even degree with constant term one. Therefore, for every $M \geq 2$,

$$C \bmod M \in \mathrm{GL}_d(\mathbb{Z}/M\mathbb{Z}).$$

This finite group contains a positive power of C equal to the identity.

Lemma 6.1. *For every $M \geq 2$ there is an effectively computable $h_0 \geq 1$ such that*

$$\mathrm{Tr}(\tau^{h_0 n}) = \mathrm{tr}(C^{h_0 n}) \equiv d \pmod{M} \qquad (n \geq 1).$$

Proof. Compute successive powers of C modulo M until the identity is reached. Finiteness and invertibility ensure termination. Every multiple of the return exponent also gives the identity, whose trace is d. □

Proof of Theorem 1.2. Take the squarefree modulus

$$M_d = \prod_{p \leq 2d-3} p,$$

where the product runs over primes. Choose h_0 as in Lemma 6.1 and then a multiple h of h_0 large enough that

$$\left\lfloor \tau^h \right\rfloor > 2d - 3. \tag{6}$$

For $r = hn$, Lemma 5.1 and the trace return give

$$\lfloor \tau^r \rfloor \equiv d - \lceil E_r \rceil = R_r \pmod{M_d}, \qquad 2 \leq R_r \leq 2d - 3.$$

Choose any prime divisor p of R_r. Then $p \mid M_d$ and $p \mid \lfloor \tau^r \rfloor$. Equation (6) and monotonicity give $\lfloor \tau^r \rfloor > 2d - 3 \geq p$. This proves compositeness and the prime-divisor bound.

Effectivity includes the size step: isolate τ by rational intervals and choose a rational u with $1 < u < \tau$. Successive multiples h of h_0 can be tested until $u^h > 2d - 2$. All such tests use rational or integer arithmetic.

Finally $\mathbb{Q}(\tau^h) = \mathbb{Q}(\tau)$. Any embedding of $\mathbb{Q}(\tau)$ fixing τ^h sends τ to a conjugate whose hth power is τ^h, hence whose modulus is τ. The only conjugate with that modulus is τ. Separability therefore forces the relative extension to have degree one. Its conjugates are the hth powers of the original conjugates, so τ^h is a Salem number of the same degree. □

Remark 6.2 (Bounds and computation). The construction is effective but is not offered as an efficient exponent-minimization algorithm. Because M_d is squarefree, one may work modulo its prime factors separately. For example, the integer

$$\mathrm{lcm}_{p \mid M_d} |\,\mathrm{GL}_d(\mathbb{F}_p)|, \qquad |\,\mathrm{GL}_d(\mathbb{F}_p)| = \prod_{i=0}^{d-1} (p^d - p^i),$$

is a valid universal matrix-return exponent for that degree and modulus. A much smaller trace-return exponent may suffice even when the matrix itself has not returned to the identity.

7 An exact quartic certificate

Consider

$$P(X) = X^4 - X^3 - X^2 - X + 1.$$

Modulo 2 it becomes $\Phi_5(X)$, which is irreducible because the order of 2 modulo 5 is 4. Thus P is irreducible over $\mathbb{Q}$. The substitution $y = X + X^{-1}$ gives

$$X^{-2}P(X) = y^2 - y - 3, \qquad y_\pm = \frac{1 \pm \sqrt{13}}{2}.$$

Here $y_+ > 2$ and $-2 < y_- < 2$. The roots associated with y_+ are a positive reciprocal pair τ, τ^{-1} with $\tau > 1$, and those associated with y_- are nonreal of modulus one. Therefore τ is a quartic Salem number.

Take the companion convention

$$C = \begin{pmatrix} 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{pmatrix}.$$

Repeated squaring in $\mathbb{Z}/30\mathbb{Z}$ yields

$$C^{260} \equiv \begin{pmatrix} 21 & 10 & 20 & 10 \\ 20 & 11 & 20 & 10 \\ 20 & 10 & 21 & 10 \\ 20 & 10 & 20 & 11 \end{pmatrix} = I + N \pmod{30}. \tag{7}$$

Direct multiplication gives $N^2 \equiv 0$ and $\operatorname{tr} N \equiv 0 \pmod{30}$. Thus

$$C^{260n} \equiv I + nN \pmod{30}, \qquad \operatorname{Tr}(\tau^{260n}) \equiv 4 \pmod{30}$$

for every $n \geq 1$. Lemma 5.1 leaves only the four residual integers $2, 3, 4, 5$, covered by the primes $2, 3, 5$.

To verify the size condition without decimal approximations, note that $y_+ > 13/6 = 3/2 + 2/3$, whence $\tau > 3/2$. Therefore $\tau^{260} > \tau^6 > (3/2)^6 = 729/64 > 11$, so its floor is at least 11 and exceeds every covering prime. We conclude that

$$\left\lfloor (\tau^{260})^n \right\rfloor \text{ is composite for every } n \geq 1.$$

The exponent 260 in this certificate is a trace-return exponent; (7) does not assert $C^{260} \equiv I \pmod{30}$.

8 A corollary for projective K3 dynamics

For an automorphism F of a compact complex surface, write $\lambda(F)$ for its first dynamical degree. The basic iteration identity is

$$\lambda(F^r) = \lambda(F)^r \qquad (r \geq 1).$$

We invoke the following existing geometric theorem, without reproving its lattice-theoretic content.

Theorem 8.1 (Brandhorst [2], Theorem 1.1)**.** *If τ is a Salem number of degree d, then some positive power of τ is the dynamical degree of an automorphism of a complex projective K3 surface if and only if $d \leq 20$.*

Corollary 8.2. *For every Salem number τ of degree at most 20, there are a positive integer H, a complex projective K3 surface X, and an automorphism G of X such that*

$$\lambda(G) = \tau^H, \qquad \lfloor \lambda(G)^n \rfloor \text{ is composite for every } n \geq 1.$$

Proof. By Theorem 8.1, choose $m \geq 1$ and F on a projective K3 surface with $\lambda(F) = \tau^m$. Choose h from Theorem 1.2, put $H = \operatorname{lcm}(m, h)$, and set $G = F^{H/m}$. Then $\lambda(G) = \tau^H$. Since H is a multiple of h, every exponent Hn belongs to the sequence already covered by Theorem 1.2. □

This conclusion is a combination of two theorems through iteration. It does not decide whether τ itself is realizable, and it does not assert compositeness of numbers of fixed points or periodic points. Those geometric counts are not, in general, the floor of a power of the dynamical degree.

9 Scope, dependencies, and outstanding questions

The two principal arguments share the finite-cover criterion but use different sources of error control. In the cubic setting the conjugate error decays, and the discriminant fixes a short return exponent. In the Salem setting the error need not decay; an integer norm removes the dangerous endpoint, while a larger finite modulus permits several possible prime divisors.

Feature	Pure cubic units	Salem units
Exponent	$6n$ for every $n \geq 1$	hn for effective h
Trace modulus	6	$\prod_{p \leq 2d-3} p$
Correction	ceiling in $\{0, 1\}$	ceiling in $\{-d+3, \ldots, d-2\}$
Prime cover	$\{2, 3\}$	primes at most $2d-3$
Size input	elementary integer norm	sufficiently large multiple of h_0

These theorems do not prove a degree-independent lower bound for Salem numbers. The exponent h can grow with the degree and with the arithmetic of the minimal polynomial. Nor do they settle a prime-generating constant's transcendence, the full cubic Pisot sparse-exponent problem, or a classification of K3 dynamical degrees.

The effective Salem construction raises concrete questions: how small can its prime cover be; when can a trace return be substantially shorter than a matrix return; and for which Salem numbers does one prime alone cover all floors along a power subsequence? These are distinct from asserting that the exponent can be bounded uniformly over all degrees.

Proof-status note 9.1. This manuscript develops source blocks T54 and T60 of the internal research audit. The local and norm arguments above are supplied in full; the K3 corollary additionally uses Brandhorst's published theorem. The quartic matrix identity is a finite certificate and the displayed cubic values are illustrations. The earlier survey-style source used the least-Pisot-number theorem for the cubic size bound; Lemma 4.1 replaces that input by an elementary proof. Historical novelty remains a separate bibliographic question and is not inferred from an unsuccessful search.

References

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